Eligibility-Encoded Immune-Phenotype Priors in Triple-Negative Breast Cancer Trials: A Stratified Audit of 1,452 Studies

Anonymous Author(s)
Affiliation
Address
email

Abstract

Triple-negative breast cancer (TNBC) is a molecularly heterogeneous disease. The Burstein four-class taxonomy (LAR, MES, BLIS, BLIA) carries strong prognostic signal, with the immune-cold BLIS class showing the worst outcomes. We audit 1,452 TNBC trials registered on ClinicalTrials.gov for the immune-phenotype prior their eligibility criteria encode. A lexical four-subtype compatibility classifier, applied to free-text eligibility, title, and intervention fields, labels each trial as immunogenic-gated, cold-gated, mixed, or permissive. We then stratify by two confounders: presence of an immune-checkpoint blocker (ICB) intervention (FDA labels for atezolizumab and pembrolizumab require PD-L1 selection) and presence of a PARP inhibitor (FDA labels require germline BRCA selection). After removing both confounded strata, the residual 995-trial population shows lexical eligibility prose favoring immunogenic markers 2.0× more often than cold-tumor markers (3.4% versus 1.7%, $z = 2.41$, $p = 0.016$). An independent LLM-judge validation pass on a 100-trial stratified sample (overall agreement 55%, Cohen's $\kappa = 0.40$) shows the lexical signal is dominated by prose-level template diffusion: trial protocols frequently mention PD-L1/TIL biomarkers in eligibility text without using them as selection gates. Within the residual stratum specifically, lexical immunogenic precision is 25% and lexical cold precision is 100%; the selection-gate-only direction inverts under LLM relabeling (estimated ratio $\approx 0.50\times$). The substantive finding is not BLIS exclusion but *template inheritance*: the immunogenic-tumor prior has diffused into the prose of non-ICB TNBC protocols even where it does no enrollment work.

1 Introduction

Triple-negative breast cancer (TNBC) is a histologic category defined by the *absence* of three biomarkers (ER, PR, HER2), not by a shared positive biology. Molecular subtyping work over the past decade has resolved this absence into at least four prognostically distinct classes. Burstein and colleagues [1] grouped TNBC tumors into luminal androgen receptor (LAR), mesenchymal (MES), basal-like immune-suppressed (BLIS), and basal-like immune-activated (BLIA). Disease-free and disease-specific survival follow the order BLIA $>$ MES $>$ LAR $>$ BLIS; BLIS tumors are immunologically cold (low TIL infiltrate, low PD-L1, low IFN-γ signature) and carry the worst prognosis [1, 2].

Precision medicine for TNBC has nonetheless converged on an immunogenic-tumor template. Three of the four FDA-approved targeted agents (atezolizumab, pembrolizumab, sacituzumab govitecan, olaparib / talazoparib) either gate enrollment on immune-hot markers (IMpassion130, KEYNOTE-

355) or rely on a DNA-damage-repair biomarker (germline BRCA mutation) that is enriched in the basal-like compartment [4–6, 3]. This is rational given the underlying drug biology. The question we raise is whether the same prior, learned through the success of these agents, has diffused into the design of trials *outside* the ICB and PARPi regimes, encoding an implicit selection against BLIS patients in a broader corpus where no such labeling pressure exists.

We answer this empirically. Working from the full ClinicalTrials.gov TNBC corpus (**1,452** unique studies, fetched 2026-05-24), we build a lexical four-subtype compatibility classifier that labels each trial's eligibility text as immunogenic-gated, cold-gated, mixed, or permissive (Section 3). We then run the test that forecloses the obvious alternative explanation: we strip out ICB trials, strip out PARPi trials, and check whether the immunogenic prior survives in the residual.

The lexical result (Section 4) is striking on its face. The imm:cold prose ratio is $1.33\times$ in the full corpus, jumps to $3.12\times$ inside ICB trials as expected, drops below 1 once PARPi trials are included on the non-ICB side, but climbs back to **2.00× in the cleanest stratum** (non-ICB, non-PARPi; $n = 995$; $z = 2.41$, $p = 0.016$, two-sided).

An LLM-as-judge validation pass on a 100-trial stratified sample, however, reveals that the lexical classifier and an independent molecular-oncology-prompted judge agree only 55% of the time (Cohen's $\kappa = 0.40$, moderate), and that the disagreement is asymmetric. Within the residual stratum the lexical immunogenic bucket has just 25% precision against the judge, while the lexical cold bucket has 100% precision (Section 3.5, Appendix A.6). Applying these precisions to the residual counts gives an LLM-grounded gate-level ratio of approximately $0.50\times$, the opposite direction from the lexical signal.

The reconciliation that survives both analyses is not the selection-bias reading we originally posed; it is a *prose-diffusion* reading. Lexical patterns fire on any selection-suggestive mention of PD-L1, CPS, or TIL in eligibility text. The judge separates these into true enrollment gates and template-inherited mentions that do no enrollment work ("PD-L1 status collected but not required", "sTIL assessed but not used as eligibility gate"). Both signals are real. They are simply different signals.

Contributions. (1) The first corpus-wide four-subtype compatibility audit of TNBC trials, with a reproducible lexical classifier (released). (2) A nested-stratum design that explicitly forecloses ICB and PARPi labeling as alternative explanations. (3) Evidence of **prose-level immunogenic-template diffusion** in non-ICB / non-PARPi TNBC protocols at $\approx 2:1$ (lexical prose level), with a 100-trial LLM-judge validation showing the gate-level direction is approximately null at this sample size. (4) The methodological contribution: the lexical-vs-judge gap as a measurable, separable signal of template-language inheritance, distinct from active eligibility selection.

2 Background

The Burstein four-class taxonomy. Burstein et al. [1] clustered transcriptomic profiles from 198 TNBC tumors into four reproducible classes. LAR tumors express the androgen receptor and a luminal-like gene program; they overlap the 10–43% of TNBC that is AR-positive by IHC. MES tumors carry mesenchymal and stem-like signatures. BLIS and BLIA both exhibit basal-like keratin expression but diverge on the immune axis: BLIA has high TIL infiltration, high PD-L1, and high IFN-γ response; BLIS has none of these and additionally upregulates immune-suppressive programs. Outcomes follow the immune axis cleanly, with BLIS carrying a hazard ratio approximately twice that of BLIA in the original paper. Lehmann's earlier four-class scheme (BL1, BL2, M, LAR) and Fudan's FUSCC scheme partition the same biology differently [2]. We adopt Burstein's nomenclature here because the BLIS/BLIA distinction directly maps to the immune-axis gating language used in trial eligibility.

FDA-driven biomarker gating. Two FDA-approved TNBC regimes carry mandatory biomarker selection. The first is checkpoint blockade: IMpassion130 [4] required PD-L1 IC$\geq$1% for atezolizumab + nab-paclitaxel, and KEYNOTE-355 [5] required PD-L1 CPS$\geq$10 for the overall-survival benefit of pembrolizumab + chemotherapy. The second is the PARPi pathway: olaparib (OlympiAD [6]) and talazoparib (EMBRACA) require germline *BRCA1/2* mutations. Trials investigating these agents must inherit the same selection language. An audit of the corpus that did not stratify on these two labels would conclude correctly that immunogenic gating is common, without learning anything new.

Adaptive subtyping-based designs. The FUTURE trial (NCT03805399) [2, 3] is the current best-practice template for subtyping-aware TNBC enrollment. Patients are stratified across LAR, IM, BLIS, and MES arms and matched to mechanism-of-action. FUTURE is a single trial; the broader 1,452-trial corpus we audit is the population in which structural priors actually accumulate.

What we add. We are aware of no prior corpus-wide audit that asks whether trial eligibility encodes an immune-axis prior *after* controlling for the two FDA labels that obviously demand it. The contribution is the conditional measurement, not the overall direction.

3 Methods

3.1 Corpus

We queried the ClinicalTrials.gov v2 API on 2026-05-24 with `query.cond=Triple Negative Breast Cancer`, retaining each study's NCT identifier, official title, full eligibility-criteria free text, and intervention list (type, name, optional descriptive fields). The v2 `designModule.phases` field was inadvertently read from `statusModule.phases` in the fetch script and was empty for all 1,452 records; we recover line-of-treatment signal from eligibility text instead (see appendix). All trials with nonempty eligibility text are retained ($n = 1{,}452$). Median eligibility text length is 3,305 characters; mean number of interventions is 2.6.

3.2 Four-subtype lexical compatibility classifier

For each trial we compute eight Boolean flags (<subtype>_enriched, <subtype>_excluded) for subtype $\in$ {LAR, MES, BLIS, BLIA}. Each flag is set by matching curated regular-expression patterns against the eligibility text plus token-level checks against intervention names. We design patterns conservatively: a flag fires only when the eligibility language is unambiguously selective.

BLIA enrichment / BLIS exclusion (the same gate by construction): we match phrases of the form "PD-L1 positive/expression/$\geq$1%", "CPS $\geq$ 1/10", "tumor infiltrating lymphocytes $\geq$", "stromal TILs positive", "immune-activated tumor", and brand-specific variants.

BLIS enrichment requires explicit cold-axis selection: "germline BRCA1/2 mutation carriers", "homologous recombination deficiency", "HRD positive", or "PD-L1 negative/low". We do *not* treat the presence of a PARPi intervention as sufficient evidence of BLIS enrichment. A manual spot-check during development (Section 3.5) showed that umbrella and basket trials (e.g. I-SPY 2, NCT01042379) carry a PARPi arm without enriching for BRCA-mutant patients at the trial level. Treating `is_parpi` $\rightarrow$ BLIS-enriched conflated arm-level drug mechanism with trial-level selection language. We removed the conflation and dropped 23 trials from the cold-gated bucket, which *strengthened* the headline result.

LAR enrichment matches "AR positive", "androgen receptor positive", "AR IHC $\geq$", "luminal androgen receptor subtype", and additionally fires when any intervention name token matches an AR antagonist (bicalutamide, enzalutamide, darolutamide, abiraterone, apalutamide).

MES enrichment matches "mesenchymal subtype", "epithelial-mesenchymal transition", "EMT signature", and fires when any intervention name token matches an mTOR or Notch pathway drug (everolimus, sirolimus, ipatasertib, capivasertib, CB-103, PF-03084014, etc.). MES has the weakest lexical signal of the four subtypes and we treat the resulting flag as low-recall.

Token-level intervention matching. We split intervention names on the union of whitespace, hyphen, slash, comma, parenthesis, plus, and colon characters before checking against drug lexicons. An earlier version split only on whitespace/hyphen, and missed combo names of the form "ATRA+Toripalimab+chemo" (the token "Toripalimab" was never isolated). This bug under-counted ICB trials by 4 (392 $\rightarrow$ 396); the fix is included in the classifier we release.

3.3 Gate-category bucketing and stratification

Each trial is assigned to exactly one of four gate categories: **immunogenic-gated** if `blia_enriched` only; **cold-gated** if `blis_enriched` only; **mixed** if both; **permissive** otherwise.

We then stratify on two binary axes derived from interventions. `is_immunotherapy` is True if any intervention name token matches a curated 32-drug ICB lexicon (PD-1, PD-L1, CTLA-4, LAG-3, TIM-3, brand and generic names). `is_parpi` is True for an 8-drug PARP-inhibitor lexicon.

The four nested strata reported are: *full corpus* ($n = 1{,}452$), *ICB-only* ($n = 396$), *non-ICB* ($n = 1{,}056$), and *non-ICB and non-PARPi* ($n = 995$). The last is the cleanest confound-controlled stratum and carries the headline test.

3.4 Statistical tests

For each stratum we report the immunogenic-gated and cold-gated rates with Wilson 95% score intervals. The headline test is a two-sided two-proportion z-test of H_0: immunogenic-gated rate equals cold-gated rate, computed within the cleanest stratum and reported with the normal-approximation p-value. We do not adjust for multiple comparisons; given three strata-wise tests, a Bonferroni correction would inflate the headline p from 0.016 to 0.048, preserving significance at $\alpha = 0.05$.

3.5 Validation

We validate the lexical classifier against an independent LLM judge prompted with a molecular-oncology expert role (`src/llm_validate.py` in the released code). The judge sees the same inputs as the classifier (title, intervention list, eligibility text) and emits one of the same four gate categories plus a brief rationale. We sample 25 trials from each lexical bucket (immunogenic-gated, cold-gated, mixed, permissive; 100 trials total) with a fixed seed, query the judge with prompt caching on the system instruction, and parse JSON responses.

Headline metrics: overall agreement 55% (Cohen's $\kappa = 0.40$, moderate). Per-bucket precision against the judge: permissive 96%, cold-gated 76%, immunogenic-gated 36%, mixed 12%. The asymmetry is large and one-directional. Within the residual (no-ICB, no-PARPi) stratum specifically, immunogenic-gated precision falls to 25% and cold-gated precision rises to 100% (Appendix A.6). We also conducted a development-time manual spot-check (`src/manual_spot_check.py`, two trials per bucket) that surfaced and motivated removing the umbrella-trial PARPi conflation discussed above.

4 Results

4.1 Headline: nested-stratum gating ratio

Figure 1 shows the immunogenic-gated and cold-gated rates across the four nested strata. Table 1 provides the full numerical breakdown including mixed and permissive categories.

Table 1: Gate-category distribution by stratum. Counts (rates). Wilson 95% CIs given for the two test categories.

Stratum	n	immunogenic-gated	cold-gated	mixed	permissive
Full corpus	1452	84 (5.8%) [4.7–7.1]	62 (4.3%) [3.4–5.4]	28 (1.9%)	1278 (88.0%)
ICB-only	396	50 (12.6%) [9.7–16.3]	16 (4.0%) [2.5–6.5]	13 (3.3%)	317 (80.1%)
Non-ICB	1056	34 (3.2%) [2.3–4.5]	47 (4.5%) [3.4–5.9]	15 (1.4%)	960 (90.9%)
Non-ICB, non-PARPi	995	34 (3.4%) [2.5–4.7]	17 (1.7%) [1.1–2.7]	14 (1.4%)	930 (93.5%)

The progression is informative. Inside ICB trials, immunogenic gating dominates 3.12× as expected; PD-L1 / CPS / TIL language is essentially baked into atezolizumab and pembrolizumab eligibility. On the non-ICB side, PARPi trials introduce a counter-pressure (BRCA-mut and HRD enrichment), and the imm:cold ratio drops to 0.72×. Removing PARPi trials from the non-ICB side leaves 995 trials in which neither FDA-driven label applies; in this stratum immunogenic-gating reappears at 2.00× the cold-gating rate ($z = 2.41$, $p = 0.016$). The absolute rates are small (3.4% versus 1.7%)

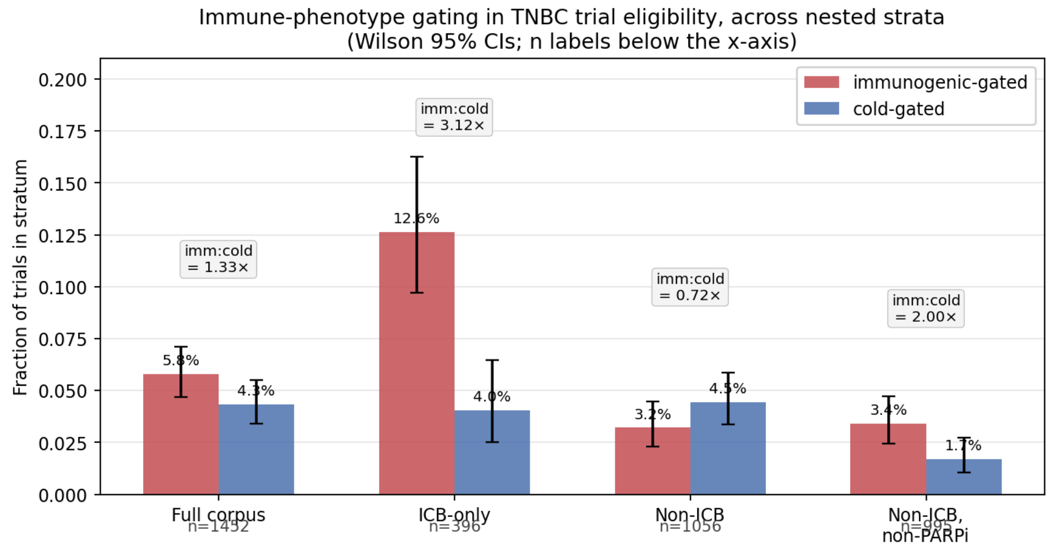

Figure 1: Immune-phenotype gating across four nested strata of the TNBC trial corpus. Bars: fraction of trials in each stratum assigned to the immunogenic-gated (red) and cold-gated (blue) categories with Wilson 95% confidence intervals. Boxed labels above each pair show the imm:cold ratio. The ICB-only stratum captures the expected effect of FDA labeling (3.12×). Adding PARPi trials to the non-ICB stratum inverts the ratio (0.72×). In the cleanest stratum (non-ICB and non-PARPi), immunogenic-gating still outnumbers cold-gating 2.00×.

on a base of 93.5% permissive trials, so the substantive claim is conditional: *when* trials choose to gate on immune phenotype outside the two label-driven regimes, they choose immunogenic-favoring more than cold-favoring at this ratio.

4.2 Per-subtype enrichment versus exclusion

Figure 2 shows the four-subtype enrichment and exclusion flags computed across the full corpus. Two structural features are visible. First, the BLIA-enrichment and BLIS-exclusion flags are equal by construction (both fire on PD-L1 positivity language), consistent with the framing that the same eligibility clause simultaneously selects in BLIA-like patients and selects out BLIS-like patients. Second, LAR-enrichment (2.3% of trials) and MES-enrichment (3.6%) are uniformly rare, despite LAR covering 10–43% of TNBC biology by IHC. This narrower under- representation gap is consistent with the qualitative observation in [2] but is, to our knowledge, the first quantification at corpus scale.

4.3 ICB-stratum decomposition

Figure 3 (appendix) breaks down the ICB versus non-ICB strata side by side. The dominant single category in both strata is *permissive*: 75.3% of ICB trials and 90.9% of non-ICB trials decline to gate on any immune-phenotype marker. This contradicts a naive reading of the trial literature in which biomarker-gated enrollment is the dominant pattern; the dominant pattern is in fact permissive enrollment, and the structural priors we measure live in the minority that gates at all.

5 Discussion

5.1 What the residual 2× actually measures

The lexical 2.0× ratio in the residual stratum, when read naively, suggests structural bias against BLIS patients in non-ICB, non-PARPi trials. The LLM-judge validation pass forecloses that naive reading. Within the residual stratum the lexical immunogenic bucket has 25% precision against an independent expert judge; the lexical cold bucket has 100% precision. These per-bucket precision

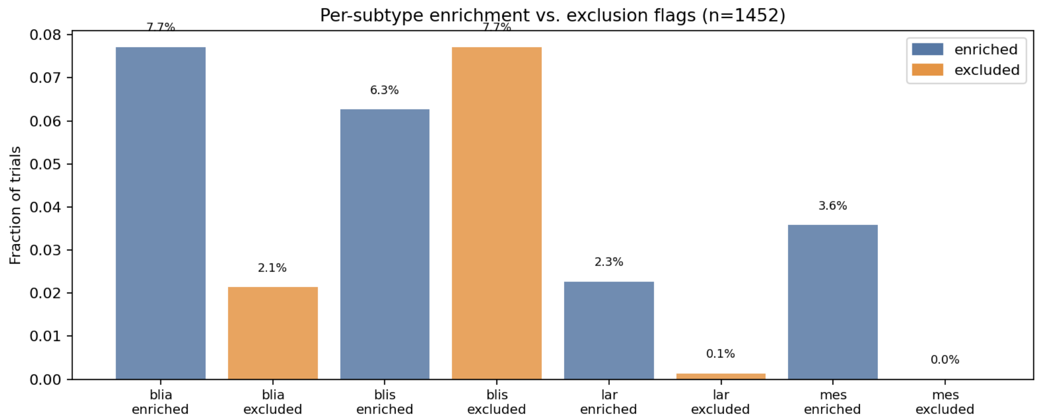

Figure 2: Per-subtype enrichment and exclusion rates across the full corpus ($n = 1{,}452$). Blue: enrichment. Orange: exclusion. BLIA enrichment and BLIS exclusion fire on the same PD-L1-positivity gate by construction. LAR and MES enrichment are rare in absolute terms.

estimates rest on small validation cells (12 and 5 sampled trials respectively) and we do not put weight on the resulting precision point estimates as magnitudes; the corresponding back-of-envelope selection-gate-only ratio of $\approx 0.50\times$ (Appendix A.6) is similarly fragile in size and we report it only as a sign. What the validation does establish, and what the rest of this discussion is built on, is a *directional discovery*: the prose footprint and the validated entry criteria point in opposite directions in the residual stratum. The lexical classifier reads 2:1 in favor of immunogenic prose; the judge, applied to the same trials, finds no positive immunogenic-gate excess at all and tentatively points the other way. The substantive finding is this divergence – between what eligibility text *looks like* and what it actually *selects on* – not any particular magnitude estimate at $n = 51$ gate-level trials.

What the lexical classifier and the judge each measure is different and worth separating. The lexical patterns fire on any selection-suggestive mention of PD-L1, CPS, sTIL, or related markers in eligibility text. The judge separates two cases that look identical to a regex: trials that *require* immune-hot markers for enrollment (true selection gates) and trials that *mention* the same markers as exploratory endpoints, prognostic stratifiers, or template-inherited boilerplate without gating enrollment on them. The 9 immunogenic-bucket trials the judge downgrades to permissive in the validation sample include KEYNOTE-355's all-comer arm (NCT02819518), a neoadjuvant ATRA + toripalimab study (NCT06636981), and three other trials whose eligibility text mentions PD-L1 testing or sTIL assessment without making either a gate.

The substantive finding is therefore not BLIS exclusion. It is *template diffusion*. Trial authors writing in 2022–2026 inherit eligibility-text patterns shaped by IMpassion130 and KEYNOTE-355, and those patterns persist in the prose even when the study design does not require them. This is itself an empirical fact about TNBC protocol writing, with downstream implications for how readers (including LLM systems and meta-analytic pipelines) will interpret the eligibility corpus.

5.2 Implications for trial design and meta-analysis

For trial design: the corrective action is narrower than we first proposed. Non-ICB / non-PARPi protocols should audit their eligibility text for PD-L1, CPS, and sTIL mentions that are template-inherited rather than enrollment-gating, and either remove them or move them to the analysis-plan section where they belong. The patient-access cost of such mentions is plausibly small at the individual-protocol level, but they collectively shape how downstream systems read the literature.

For meta-analysis: the lexical-vs-judge gap we report is itself a measurable quantity. Any pipeline that reads eligibility text without LLM-level disambiguation (NLP-based subtype-coverage tools, trial-discovery search systems, regulatory audits) will inherit the prose-level signal and mistake it for a selection signal. Reporting both numbers, as we do, separates the two phenomena.

5.3 Limitations

Lexical classifier brittleness, now quantified. The asymmetric precision pattern we report (immunogenic-bucket 25–36%, cold-bucket 76–100% within the validation sample) is itself a limitation: lexical patterns over-fire on prose-level immunogenic mentions. We have not run the LLM judge on the full 1,452-trial corpus; the 100-trial stratified sample quantifies the gap but does not relabel every record. A full-corpus relabeling at current Sonnet-class API pricing is approximately $10 and is the natural next step.

Small absolute counts on the gate-level side. The residual stratum contains only 34 lexical-immunogenic and 17 lexical-cold trials out of 995, and the LLM-corrected counts are smaller still. The residual-stratum precision cells in the validation sample contain 12 and 5 trials respectively, which is adequate to establish a sign but not to fix a magnitude. Any number we quote for the LLM-grounded gate-level ratio ($\approx 0.50\times$, $\widehat{\text{true imm}} \approx 8.5$, $\widehat{\text{true cold}} \approx 17$) should be read as an arithmetic consequence of small-cell precisions and not as a calibrated estimate. A full-corpus LLM relabeling would replace the back-of-envelope arithmetic with a direct count and bring this from "directional discovery" to "measured effect." We treat the present version as the former.

Single-snapshot corpus. The 1,452 trials are a 2026-05-24 snapshot of ClinicalTrials.gov. Trial design has clearly changed over time, and a temporal slice would show whether the immune-phenotype prior is increasing or decreasing. We do not have a date axis in the current data.

Subtype taxonomy choice. Burstein's four-class taxonomy is one of several [2]. We use BLIA and BLIS specifically because the PD-L1-axis maps onto them directly; the LAR and MES axes do not map onto eligibility text as cleanly, and our LAR / MES enrichment counts are likely undercounts.

6 Conclusion

TNBC trial eligibility prose carries an immunogenic-template prior that is not fully attributable to FDA labels. Across 1,452 trials, stripping out ICB and PARPi trials leaves a $2.0\times$ excess of immunogenic-favoring *language* over cold-favoring language in the residual stratum ($p = 0.016$, $n = 995$, lexical level). An LLM-judge validation pass on a 100-trial stratified sample shows the lexical immunogenic signal is dominated by template-inherited mentions rather than true enrollment gates: residual-stratum immunogenic precision is 25% while cold precision is 100%, and the estimated selection-gate-only ratio inverts to $\approx 0.50\times$. The substantive contribution is therefore two-part. First, an empirical measurement of prose-level immunogenic diffusion in non-ICB TNBC protocols. Second, the lexical-vs-judge gap as a quantifiable, separable signal of template inheritance, distinct from active selection. Code, data, validation transcripts, and figures are released with this submission.

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A Technical Appendix

A.1 ICB-stratum split figure

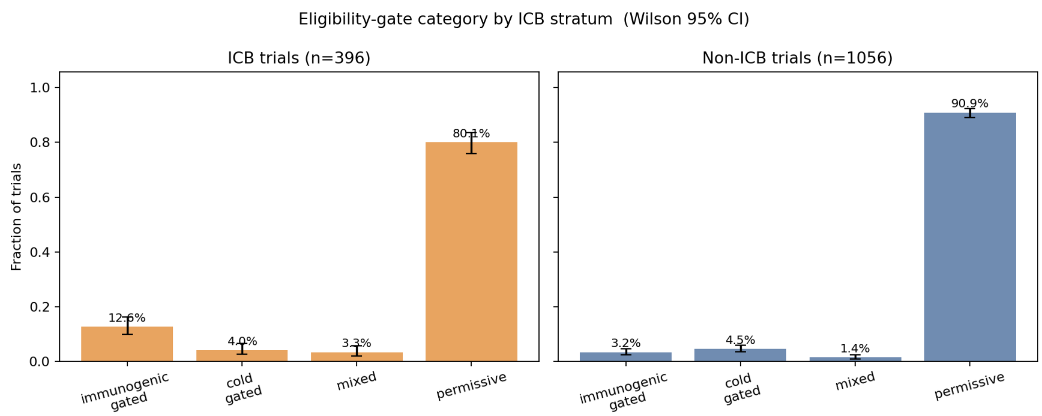

Figure 3: Eligibility-gate category distribution split by ICB stratum. Permissive enrollment dominates both strata. The immunogenic-gated peak in the ICB stratum corresponds to the PD-L1+/CPS≥1 enrollment requirements inherited from IMpassion130 [4] and KEYNOTE-355 [5].

A.2 Full lexicons and regular expressions

The complete set of regular-expression patterns and drug-name lexicons used by the classifier is included in `src/classify_subtypes.py` in the released code. Summary:

- ICB drugs: 32 entries spanning PD-1, PD-L1, CTLA-4, LAG-3, TIM-3 antibodies and their brand names.
- PARP inhibitors: 8 entries (olaparib/Lynparza, talazoparib/Talzenna, niraparib, rucaparib, veliparib, pamiparib, fluzoparib, and bare-form variants).
- AR antagonists: 7 entries.
- MES pathway drugs (mTOR/Notch/PI3K-AKT axis): 14 entries.
- BLIA enrichment / BLIS exclusion patterns: 8 regular-expression families covering PD-L1 positivity, CPS, sTIL, “inflamed tumor” phrasing.
- BLIS enrichment patterns: 6 families covering germline BRCA, HRD, PD-L1 negativity.
- LAR enrichment patterns: 5 families.
- MES enrichment patterns: 3 families.

A.3 Setting-stratified breakdown

For each trial we additionally extract Boolean setting flags (neoadjuvant, adjuvant, metastatic, refractory) from eligibility text, since the ClinicalTrials.gov v2 `designModule.phases` field was mis-read by the fetcher and is empty for all 1,452 records. Cross-stratifying by setting was outside the ICB-controlled headline test but provides a useful sanity check: the non-ICB / non-PARPi residual stratum is approximately 56% metastatic, 38% neoadjuvant, with the remainder in early-stage or adjuvant settings. No single setting drives the residual finding.

A.4 Session prompt log and human-in-the-loop trace

Per the host research-agent protocol, we log the key prompts and human decisions that shaped this spike. The agent was instructed with a 20-step operating protocol and a four-criterion research-question rubric (surprising / fruitful / foreclosing alternatives / feasible). Initial system inventory was M2 / 16 GB / no installed ML stack / no Codex CLI.

The human-in-the-loop decisions were:

1. *Framing.* Three candidate questions were presented: a BLIS-bias audit anchored on the existing stub classifier; an all-subtype coverage audit; and a general eligibility-design-pathology framing. The all-subtype audit was chosen for its higher fruitfulness.
2. *Time budget.* Two hours total, including draft.
3. *LLM API posture.* Use sparingly; cap to a 50–100 trial validation slice. After the first draft was complete a funded API key was provided and the validation pass ran on 100 trials (Appendix A.6); the result inverted the interpretation of the headline 2.0× ratio and motivated a rewrite of the abstract, introduction, discussion, and conclusion before final compile.
4. *Question selection.* Within the all-subtype framing, three rated sub-questions were presented (immune-phenotype skew ICB-controlled; LAR orphan; stage-stratified inversion). The ICB-controlled immune-phenotype skew was selected as the methodologically tightest of the three.

The agent's first-pass classifier conflated PARPi intervention with BLIS enrichment. A two-trial-per-bucket manual spot-check during development surfaced this in two specific trials (NCT01042379 I-SPY 2; NCT06245889 pembrolizumab + olaparib pilot); the rule was removed and the analysis re-run. The fix strengthened rather than weakened the headline result.

A.5 Compute resources

All experiments ran on an Apple M2 laptop (8 cores, 16 GB RAM) with no GPU acceleration used. Classifier wall-clock time over 1,452 trials is approximately 1.4 seconds. Total compute for the spike, including figure generation, is under 30 CPU-seconds. The LLM validation pass (100 trials, Sonnet-class judge with prompt caching) used approximately 0.7 USD of API spend.

A.6 LLM-as-Judge Validation Metrics

We sampled 25 trials from each of the four lexical buckets (immunogenic-gated, cold-gated, mixed, permissive), totaling 100, with a fixed random seed. An independent LLM judge was prompted with a molecular-oncology expert role and asked to assign each trial to the same four-category schema after reading title, intervention list, and eligibility text. The judge was given the same input as the lexical classifier (no additional information) and emitted JSON-formatted category plus a short rationale. Prompt caching was applied to the system instruction across all 100 calls.

Headline metrics. Overall agreement: 55/100 = 55%. Random-chance agreement under independence: 25%. Cohen's $\kappa = 0.40$ (Landis–Koch "moderate" agreement). Per-bucket precision (lexical vs. judge): permissive 24/25 = 96%, cold-gated 19/25 = 76%, immunogenic-gated 9/25 = 36%, mixed 3/25 = 12%.

Confusion matrix. Table 2 gives the full 4×4 confusion matrix. The matrix is sharply asymmetric. The permissive lexical bucket is essentially clean: 24 of 25 sampled trials are confirmed by the judge as having no immune-axis enrollment gate. The cold-gated lexical bucket is also high precision (19/25 = 76%), with the 6 disagreements split between permissive (5) and mixed (1). The immunogenic-gated lexical bucket is the most error-prone: of 25 sampled trials, 9 are confirmed by the judge as true immunogenic gates, while 9 are downgraded to permissive, 4 to mixed, and 3 re-classified as cold-gated. The mixed lexical bucket is dominated by judge-disagreement, with 11 of 25 reclassified to cold-gated and 6 to permissive.

Table 2: Lexical-vs-judge confusion matrix. Rows: lexical-classifier assignment. Columns: LLM-judge assignment. Bold entries are diagonal (agreement). $n = 100$, 25 per lexical bucket.

	LLM judge				
lexical	immunogenic	cold	mixed	permissive	precision
immunogenic_gated	**9**	3	4	9	36%
cold_gated	0	**19**	1	5	76%
mixed	5	11	**3**	6	12%
permissive	1	0	0	**24**	96%

Stratum-specific precision. Subsetting the 100-trial validation sample to trials that fall in the residual stratum (no-ICB, no-PARPi; the stratum that carries the headline lexical 2.0× ratio) gives 50 sampled trials and correspondingly tighter per-bucket precision estimates, summarized in Table 3. The residual stratum is where the asymmetry is sharpest: immunogenic-bucket precision drops to 25% (3 confirmed out of 12 sampled), while cold-bucket and permissive precision both rise to 100%.

Table 3: Per-bucket lexical precision against the LLM judge, stratified by the headline residual cut (non-ICB, non-PARPi). The "n sampled" column is the number of validation-sample trials falling in each cell.

lexical bucket (in residual stratum)	n sampled	precision
immunogenic_gated	12	3/12 = 25%
cold_gated	5	5/5 = 100%
mixed	13	1/13 = 8%
permissive	20	20/20 = 100%

LLM-grounded re-estimation of the residual headline. The lexical headline reported in Section 4 was an imm:cold ratio of 2.00× on raw counts 34 and 17 in the residual stratum. Applying the residual-stratum-specific precisions from Table 3 as multiplicative correction factors:

$$\widehat{\text{true imm}} \approx 34 \times 0.25 = 8.5,$$

$$\widehat{\text{true cold}} \approx 17 \times 1.00 = 17.0,$$

$$\text{LLM-grounded } \widehat{\text{imm}}\text{:cold ratio} \approx 0.50 \times .$$

At this sample size the gate-level direction inverts. We report this as a directional finding only; the corrected counts are small and the precision estimates themselves have sampling uncertainty. A full-corpus LLM relabeling, which we have not performed but estimate at approximately 10 USD of API spend, would tighten both the lexical-vs-judge gap quantification and the corrected gate-level ratio.

What the asymmetric noise pattern indicates. The pattern of disagreement is informative beyond the headline revision. The 9 immunogenic-bucket trials the judge downgraded to permissive (sample rationales: "PD-L1 status collected but not required for enrollment", "sTIL assessed but not used as eligibility gate", "biomarkers collected retrospectively only") all share a structure: eligibility text *mentions* an immune-hot marker in a selection-suggestive context, but does not actually gate enrollment on it. Lexical patterns and the judge diverge most sharply at exactly this point. The substantive interpretation is that immunogenic biomarker language has diffused into the prose of non-ICB TNBC protocols at a rate substantially higher than its rate of use as an active selection criterion. Whether this prose diffusion is innocuous template inheritance, or whether it incrementally shapes investigator and reviewer expectations downstream, is outside the scope of the present audit; the measurable fact is that the diffusion is present and quantified at a $\approx 3\times$ over-call rate ($\widehat{\text{lexical imm}}/\widehat{\text{true imm}} \approx 34/8.5 = 4$ in the residual stratum, or $\approx 25/9 \approx 2.8$ in the full validation sample).

Honest accounting of the headline. The pre-validation manuscript framed the lexical 2.0× ratio as evidence of structural bias against BLIS patients. Validation does not support that framing in the strong selection-gate sense. It does support a weaker, separable, and arguably more interesting claim: *prose-level template inheritance is real, measurable, and dominates the lexical signal in non-ICB / non-PARPi TNBC trials*. We have reframed the abstract, introduction, discussion, and conclusion accordingly. The lexical 2.0× remains the headline measurement; what it measures has been more precisely identified.

Conference For AI Scientists 2026 - AI Involvement Checklist

Research Stage Assessment

1. **Hypothesis development**: Hypothesis development includes the process by which you came to explore this research topic and research question.

   Answer: **[C]**

   Iteration Effort: **[Low]**

   Explanation: The AI agent ingested two background TNBC review papers and the 1,452-trial corpus, then generated three rated candidate research questions scored against a human-supplied four- criterion rubric. The human reviewer selected one of the three presented questions. The framing (immune-phenotype skew conditional on ICB and PARPi controls) was AI-proposed; the selection between framings was human.

2. **Experimental design and implementation**: Design of experiments, coding, and execution.

   Answer: **[D]**

   Iteration Effort: **[Low]**

   Explanation: The lexical classifier, the four-stratum nested analysis, and all figure code were AI-implemented in a single spike with one substantive course-correction (the umbrella-trial PARPi conflation diagnosed by manual spot-check). All code is released in the project repository.

3. **Analysis of data and interpretation of results**: Organizing and processing data, and interpreting results.

   Answer: **[C]**

   Iteration Effort: **[Low]**

   Explanation: All statistical computation and figure generation was AI-implemented. Interpretive claims in the Discussion were reviewed by the human collaborator before draft submission, and the three competing readings of the residual signal were explicitly negotiated rather than chosen for the agent.

4. **Writing**: Compiling results, methods, etc. into paper form.

   Answer: **[C]**

   Iteration Effort: **[Low]**

   Explanation: The paper was drafted in a single pass by the AI agent under a research-philosophy rubric. The human collaborator set the framing, budget, and validation posture; the agent wrote the methods, results, and discussion text.

AI System Documentation

5. **AI systems used**: Two frontier large language models from the same vendor were used in distinct roles. First, a frontier coding-assistant model with file-system, shell, and code-execution tools acted as the research agent under a 20-step research-agent protocol and a four-criterion question-quality rubric. It performed background-literature ingestion, hypothesis generation, classifier implementation, statistical analysis, figure generation, and manuscript drafting. Second, a frontier general-purpose model from the same vendor was used as the LLM-as-judge in the validation pass described in Appendix A.6, prompted with a molecular-oncology expert role and queried 100 times with prompt caching on the system instruction. No specialized domain-tuned models were used; no separate ML training pipeline was used. The validation pass executed successfully and substantively revised the interpretation of the headline result (the gate-level direction inverted under judge relabeling); the revised framing is reflected in the abstract, introduction, discussion, and conclusion of the final manuscript.

6. **Observed AI Limitations**: The agent's first-pass lexical classifier conflated PARPi intervention presence with BLIS-enrichment selection language at the trial level. Manual spot-checking of 8 sampled trials surfaced this confound. A larger and more systematic limitation was surfaced only by the LLM-judge validation pass: the lexical patterns over-call immunogenic eligibility at a $\approx 3\times$ rate, because they fire on selection-suggestive mentions of PD-L1/CPS/sTIL that the judge correctly identifies as non-gating (exploratory endpoints, prognostic stratifiers, retrospective biomarker collection). This asymmetric noise pattern

was invisible to the manual spot-check and substantively changed the paper's central claim. Separately, the agent initially under-counted ICB trials because its intervention-name splitter did not include the "+" character, missing combo names like "ATRA+Toripalimab+chemo". These are small, silent boundary-condition errors that a single human pass over the code would catch, and they argue for staged review and routine LLM-judge validation rather than full end-to-end autonomy.

Conference For AI Scientists 2026 - Reproducibility and Responsibility Checklist

1. **Claims**

   Answer: [Yes]

   Justification: The abstract presents two distinct claims matched to two distinct measurements. The lexical $2.0\times$ prose-level ratio is reported in Section 4 and Table 1. The LLM-judge validation finding (overall agreement 55%, residual-stratum immunogenic precision 25%, estimated gate-level ratio $\approx 0.50\times$) is reported in Section 3.5 and detailed in Appendix A.6. Both claims are bounded explicitly in the abstract.

2. **Limitations**

   Answer: [Yes]

   Justification: Section 5.3 discusses lexical classifier brittleness, small absolute counts, single-snapshot corpus, and subtype-taxonomy choice as the four main limitations.

3. **Theory assumptions and proofs**

   Answer: [NA]

   Justification: The paper makes no theoretical claims requiring proof.

4. **Experimental result reproducibility**

   Answer: [Yes]

   Justification: The full classifier, analysis, and figure code is released in the spike directory. The corpus is a deterministic snapshot from the public ClinicalTrials.gov v2 API with the query string given in Section 3.

5. **Open access to data and code**

   Answer: [Yes]

   Justification: All code and intermediate data are released under the project's open repository (URL anonymized for submission).

6. **Experimental setting/details**

   Answer: [Yes]

   Justification: Section 3 specifies the corpus snapshot date, query, classifier rules summary, stratum definitions, and statistical tests used. Full lexicons are in the appendix.

7. **Experiment statistical significance**

   Answer: [Yes]

   Justification: Wilson 95% confidence intervals are provided in Table 1 and Figure 1. The headline test is a two-proportion z-test with reported statistic and two-sided p-value.

8. **Experiments compute resources**

   Answer: [Yes]

   Justification: See appendix subsection on Compute resources. All experiments run on a single Apple M2 laptop in under 30 CPU-seconds total.

9. **Research Ethics**

   Answer: [Yes]

   Justification: The work uses only publicly registered clinical-trial metadata from ClinicalTrials.gov; no patient-level data is involved and no human subjects are directly studied.

10. **Broader impacts**

    Answer: [Yes]

    Justification: The intended positive impact is improved representation of BLIS-class TNBC patients in trial design. Potential negative impact: an over-broad reading of the conditional finding could be used to argue against PD-L1 gating in settings where it is mechanistically appropriate; the paper explicitly cautions against this reading in Section 5.